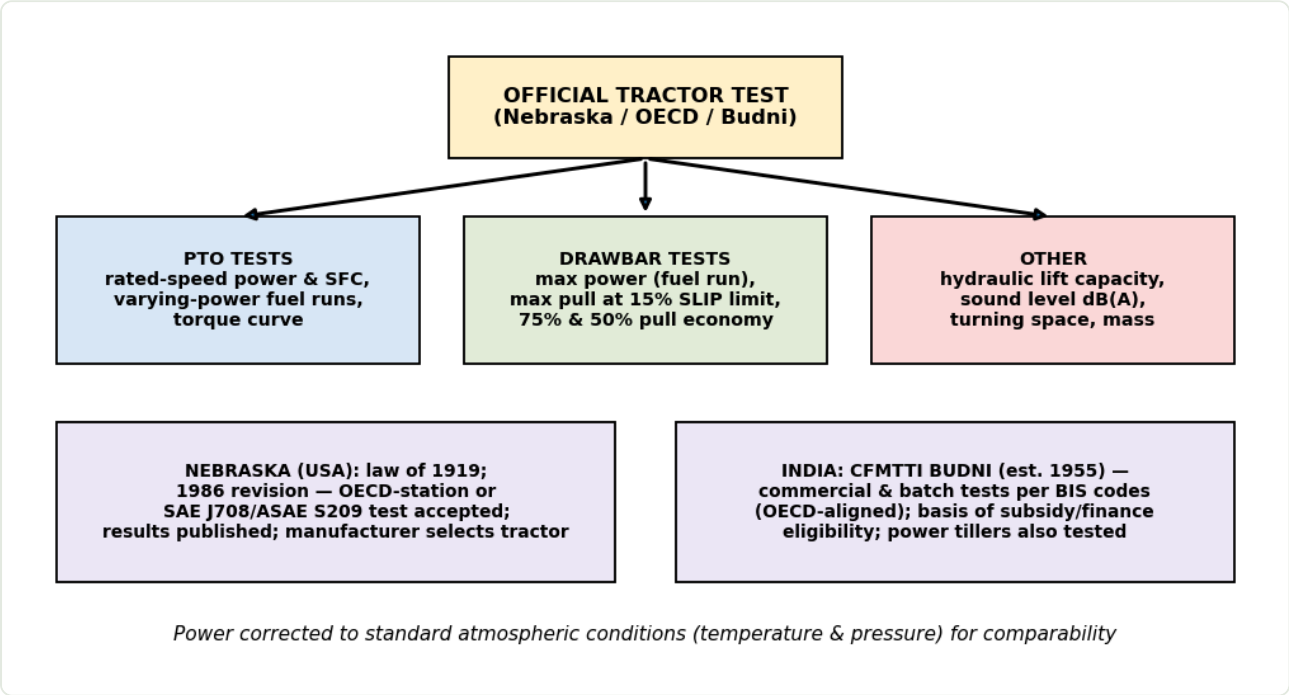
UNIT 4 · TOPIC 11

Tractor Performance Tests

Syllabus topic: Tractor performance tests

Performance criteria (Liljedahl pp. 403–405)

- The best criterion depends on intended use (p. 403).
- **Maximum drawbar power** is usually the most useful single criterion for farm tractors; for PTO-driven implements, **maximum PTO power** matters most (p. 403).
- Fuel consumption (SFC) indicates efficiency directly or indirectly (p. 404).
- Since **1971** the Nebraska tests also report **sound pressure level in dB(A)** at the operator's ear (p. 404) — the ergonomics link (Unit 5). Radial-tyre supplemental data were added later.



What an official tractor test covers (Nebraska / OECD / Budni)

The Nebraska tractor tests (pp. 420–432)

- **Origin:** in **1919** the Nebraska legislature passed a bill requiring OFFICIAL TESTS of traction engines sold in Nebraska and adequate service stations (p. 420).
- Run by the **University of Nebraska** — the world's benchmark public tractor test. Every new model sold in the state needs a test-based sales permit.

Test	Content
PTO runs	Maximum/rated speed and varying power, for fuel mapping
Drawbar runs	Maximum available power (two-hour fuel run); MAXIMUM PULL AT **15% SLIP** LIMIT; part-load economy runs
FWA (front-wheel-assist) versions	Supplemental tests without ballast redistribution (p. 431)
Constant-power (torque-rise) claims	Verified at the PTO across the claimed speed range (p. 431)

- The **MANUFACTURER SELECTS** the tractor tested — deliberately, to cut assembly-line variability (±5% power spread among same-day tractors, Fig. 14-16) and make inter-model comparisons exact (pp. 431–432).

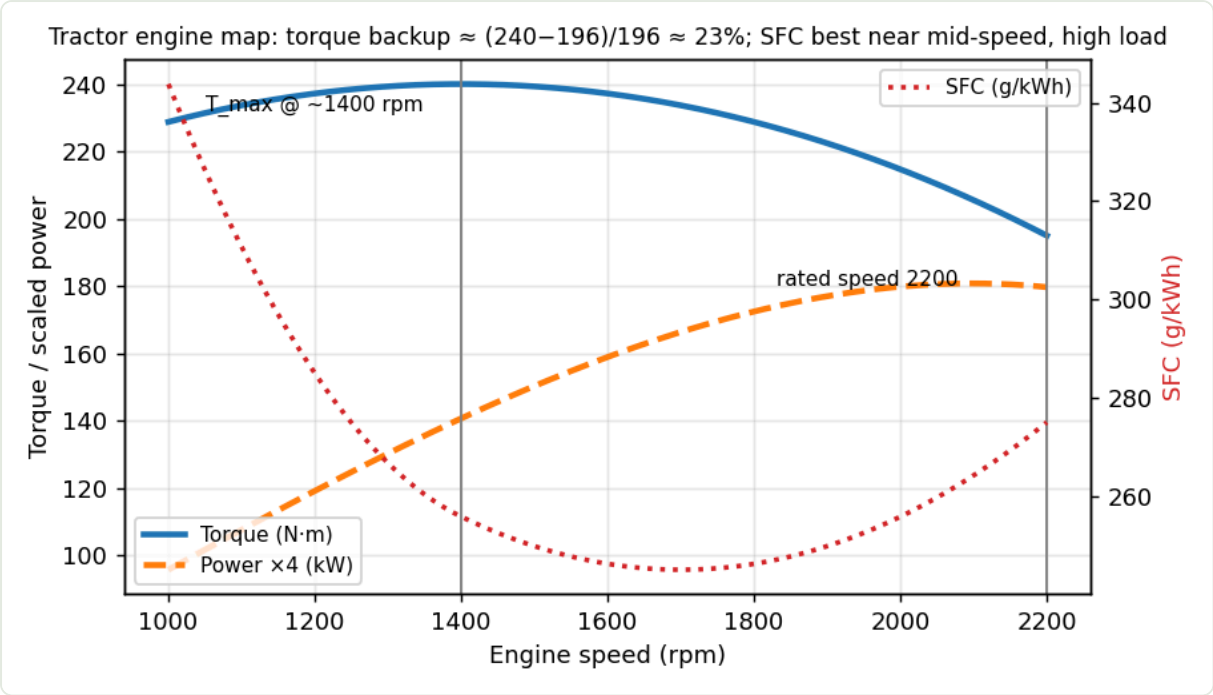
1986 revision (LB 708) (p. 432):

- Agricultural tractors only.
- Test at Nebraska OR any OECD test station, per the official OECD procedure or the SAE J708/ASAE S209 code.
- Service-parts availability declared.
- Results PUBLISHED.

OECD codes and globalization (pp. 432–433)

- OECD tractor test codes are the internationally accepted standards — 26 industrialized member nations at the book's writing; several European countries and Japan run OECD stations.
- **NIAE Silsoe** (England) conducted OECD/EEC/ISO tests (p. 433).
- OECD codes now also cover ROPS strength tests (Unit 5).
- Measured power is **CORRECTED TO STANDARD ATMOSPHERIC CONDITIONS** (temperature, pressure — hot, thin air lowers output) so results compare fairly (p. 433).

Torque curves, engine performance and reliability (pp. 435–438)



Tractor engine map — torque backup ≈ 23%; SFC best near mid-speed at high load

- Published torque curves show torque rise (backup) from rated speed to T_max at around **60–70%** of rated speed — the lugging reserve (Topic 3a).
- SFC maps show the best economy at high load near mid-speed ("gear-up, throttle-down" — Unit 3, Topic 10a).
- Tractor reliability is treated from p. 438 — linked to the MTBF concepts of Unit 3, Topic 11.

Tractor testing in India

- **CFMTTI BUDNI** (MP, est. **1955**) is India's central tractor-testing station, with NRFMTTI Hisar and others (Unit 3, Topic 2a).
- It conducts COMMERCIAL (initial and batch) tests per BIS test codes aligned to OECD; the test report governs eligibility for institutional finance and subsidy. It also tests POWER TILLERS.
- CMVR/ARAI cover road-registration aspects and TREM emission certification.

Practice MCQs

Q176. The most useful single performance criterion for farm tractors is:

(a) Engine rpm
(b) Maximum drawbar power
(c) Paint quality
(d) Mass

Q177. For a tractor mainly running PTO implements, the criterion of most interest is:

(a) Drawbar pull
(b) Maximum PTO power
(c) Turning circle
(d) Ground clearance

Q178. The Nebraska tractor test law was enacted in:

(a) 1905
(b) 1919
(c) 1947
(d) 1986

Q179. Nebraska tests are conducted by:

(a) USDA
(b) University of Nebraska
(c) SAE
(d) John Deere

Q180. Nebraska began reporting operator-ear sound level in dB(A) from:

(a) 1955
(b) 1971
(c) 1986
(d) 2000

Q181. In official tests, maximum drawbar pull is reported at a slip limit of:

(a) 5%
(b) 10%
(c) 15%
(d) 25%

Q182. The tractor tested at Nebraska is selected by:

(a) Random pick from dealers
(b) The manufacturer
(c) Farmers
(d) Lottery

Q183. That selection policy is justified because it:

(a) Hides defects
(b) Reduces assembly-line variability, making model comparisons exact
(c) Saves money
(d) Speeds testing

Q184. The 1986 Nebraska revision allows testing at:

(a) Any garage
(b) An OECD test station per OECD or SAE J708/ASAE S209 codes
(c) Farms
(d) Dealer sheds

Q185. A key public feature of Nebraska/OECD testing is:

(a) Secret results
(b) Published test results
(c) No records
(d) Verbal reports

Q186. Internationally accepted tractor test codes are issued by:

(a) WTO
(b) OECD
(c) IMF
(d) FAO

Q187. The famous UK station that carried out OECD/EEC/ISO tractor tests is at:

(a) Oxford
(b) Silsoe (NIAE)
(c) London
(d) Leeds

Q188. Measured tractor power is corrected for:

(a) Paint colour
(b) Atmospheric temperature and pressure to standard conditions
(c) Operator weight
(d) Tyre brand

Q189. Constant-power (torque-rise) claims are verified preferably at the:

(a) Drawbar
(b) PTO across the claimed speed range
(c) Fuel tank
(d) Radiator

Q190. India's central tractor testing institute is at:

(a) Hisar
(b) Budni (MP)
(c) Anantapur
(d) Bhopal

Q191. CFMTTI Budni was established in:

(a) 1947
(b) 1955
(c) 1971
(d) 1986

Q192. A favourable Budni test report matters commercially because it:

(a) Is decorative

(b) Underpins subsidy/finance eligibility of the model

(c) Sets fuel price

(d) Fixes tax

Answers

Q176 — (b) Maximum drawbar power — *Liljedahl p. 403.*

Q177 — (b) Maximum PTO power — *Liljedahl p. 403.*

Q178 — (b) 1919 — *Liljedahl p. 420 — the 1919 law.*

Q179 — (b) University of Nebraska

Q180 — (b) 1971 — *Liljedahl p. 404 — 1971, dB(A) at the operator's ear.*

Q181 — (c) 15% — *Liljedahl p. 431 — maximum pull at the 15% slip limit.*

Q182 — (b) The manufacturer

Q183 — (b) Reduces assembly-line variability, making model comparisons exact — *Liljedahl pp. 431–432, Fig. 14-16 — random same-model tractors spread ~±5% in power.*

Q184 — (b) An OECD test station per OECD or SAE J708/ASAE S209 codes — *Liljedahl p. 432 — 1986 LB 708 revision (OECD stations, SAE J708/ASAE S209, published results, parts availability).*

Q185 — (b) Published test results

Q186 — (b) OECD

Q187 — (b) Silsoe (NIAE)

Q188 — (b) Atmospheric temperature and pressure to standard conditions — *Liljedahl p. 433 — atmospheric correction.*

Q189 — (b) PTO across the claimed speed range — *Liljedahl p. 431 — constant-power claims verified at the PTO.*

Q190 — (b) Budni (MP)

Q191 — (b) 1955

Q192 — (b) Underpins subsidy/finance eligibility of the model

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units (2nd ed., 1989), Ch. 14 "Tractor Tests and Performance", pp. 403–446; MoRTH TREM notifications (as verified in the source notes, July 2026); CFMTTI Budni documentation; Jagdishwar Sahay (maintenance schedules); NPTEL lectures.